Ansys
2025 R2

Ansys Fluent Simulation Report

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Table of Contents

System Information

Application	Fluent
Settings	2d, double precision, pressure-based, SST k-omega
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:44:41 EDT
CPU	AMD Ryzen 7 6800H with Radeon Graphics
OS	Windows

Geometry and Mesh

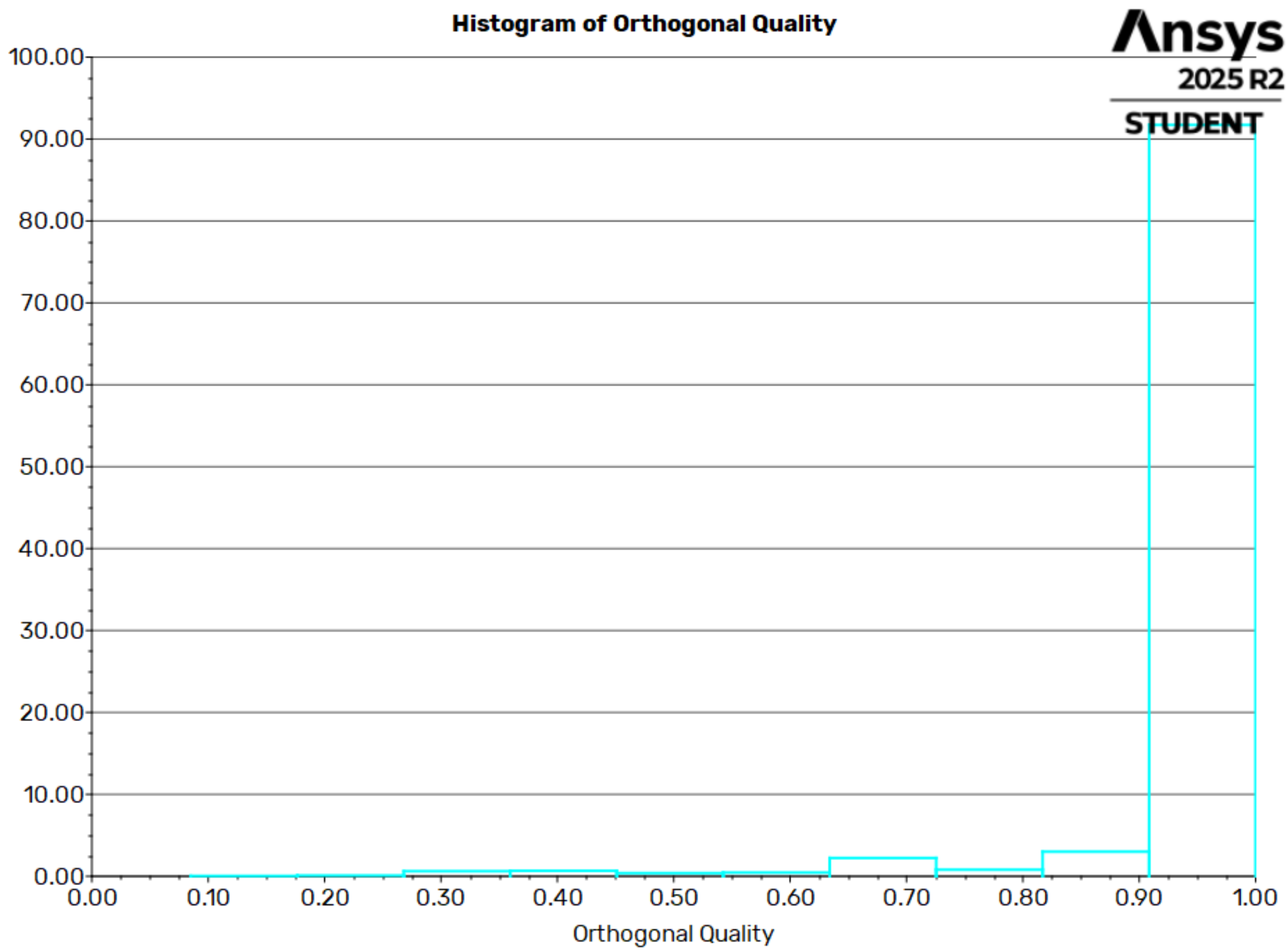
Mesh Size

Cells	Faces	Nodes
36476	59115	22612

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
surface	Mixed Cell	0.085059045	43.171374

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Steady
Viscous	SST k-omega turbulence model

Material Properties

− Fluid	
− air	
Density	1.225 kg/m^3
Viscosity	1.7894e-05 kg/(m s)
− Solid	
− aluminum	
Density	2719 kg/m^3

Cell Zone Conditions

− Fluid	
− surface	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no

Boundary Conditions

– Inlet	
– inlet	
Velocity Specification Method	Components
Reference Frame	Absolute
Supersonic/Initial Gauge Pressure [Pa]	0
X-Velocity [m/s]	-30
Y-Velocity [m/s]	-30
Turbulence Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
– Outlet	
– outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
– Wall	
– part-boundary	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0

Wall Roughness Constant	0.5
– walls	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	10 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	surface

Solver Settings

– Equations	
Flow	True
Turbulence	True
– Numerics	
Absolute Velocity Formulation	True
– Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Turbulent Kinetic Energy	0.75

Specific Dissipation Rate	0.75
Turbulent Viscosity	1
Explicit Momentum	0.5
Explicit Pressure	0.5
− Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
− Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
− Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	4
Case Read	8.763 seconds
Iteration	3.58 seconds
AMG	2.305 seconds
Virtual Current Memory	1.40772 GB
Virtual Peak Memory	1.61416 GB
Memory Per M Cell	20.9894

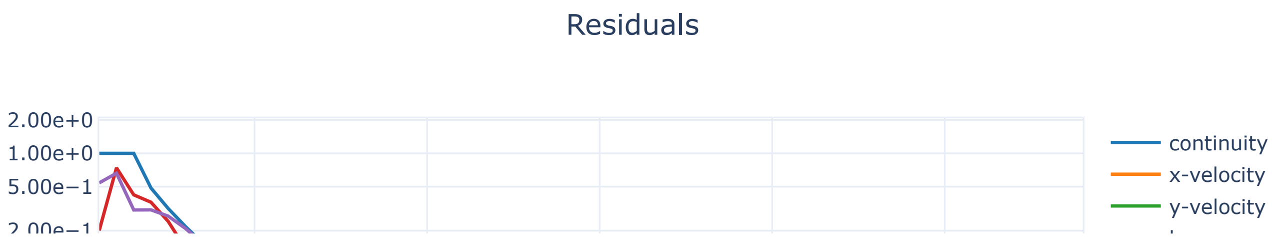
Solution Status

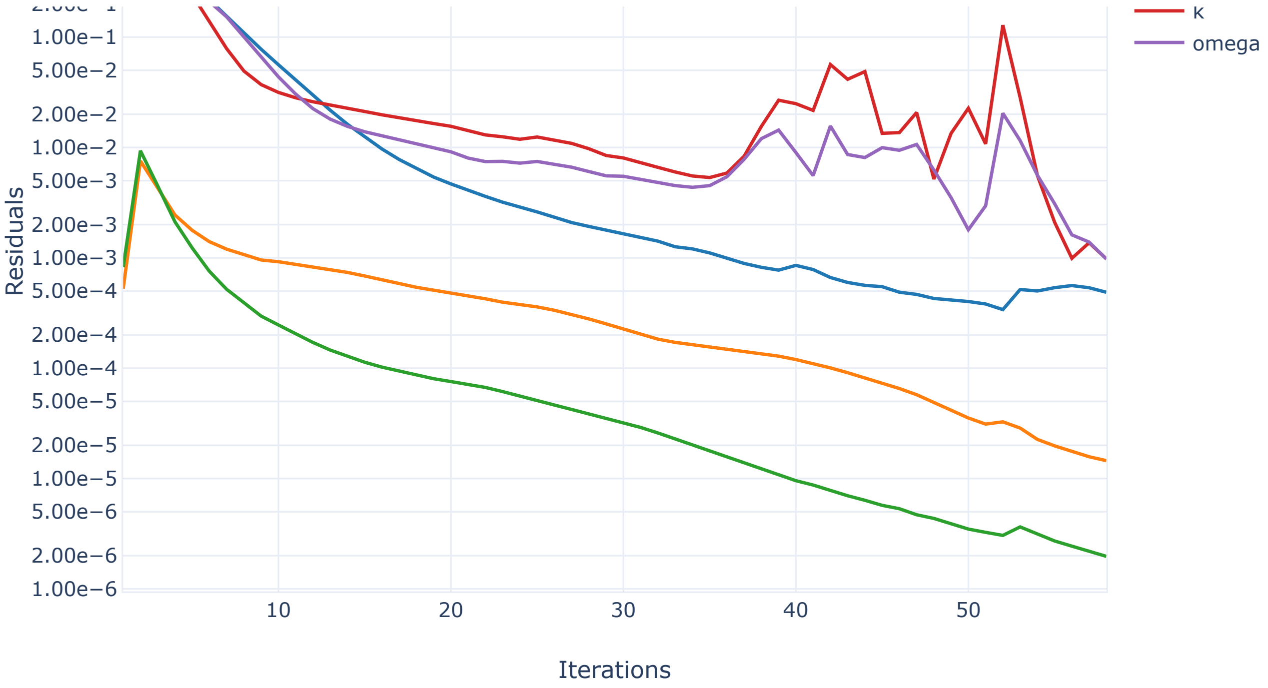
Iterations: 58

	Value	**Absolute Criteria**	**Convergence Status**
continuity	0.0004872403	0.001	Converged
x-velocity	1.451376e-05	0.001	Converged
y-velocity	1.971207e-06	0.001	Converged
k	0.0009787	0.001	Converged
omega	0.0009750085	0.001	Converged

Plots

Residuals





Contours

contour-2

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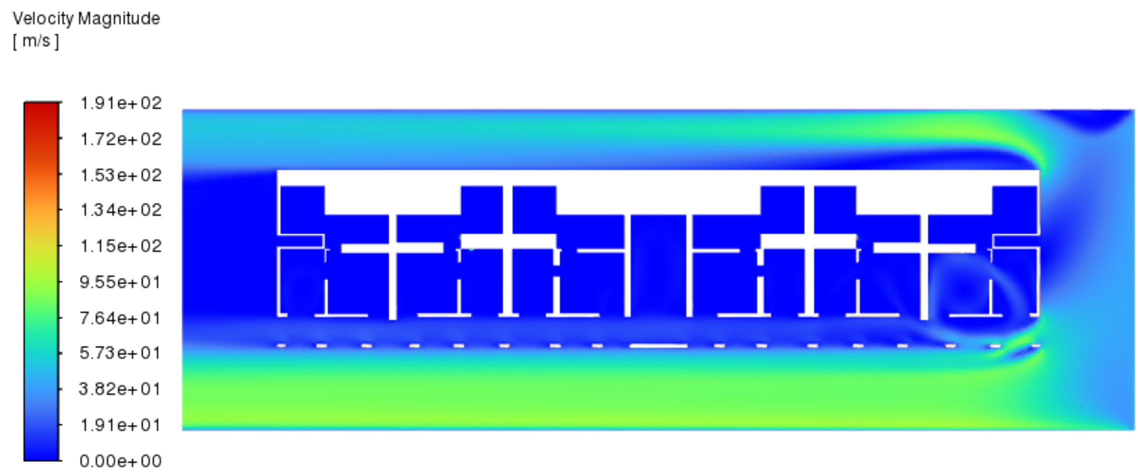
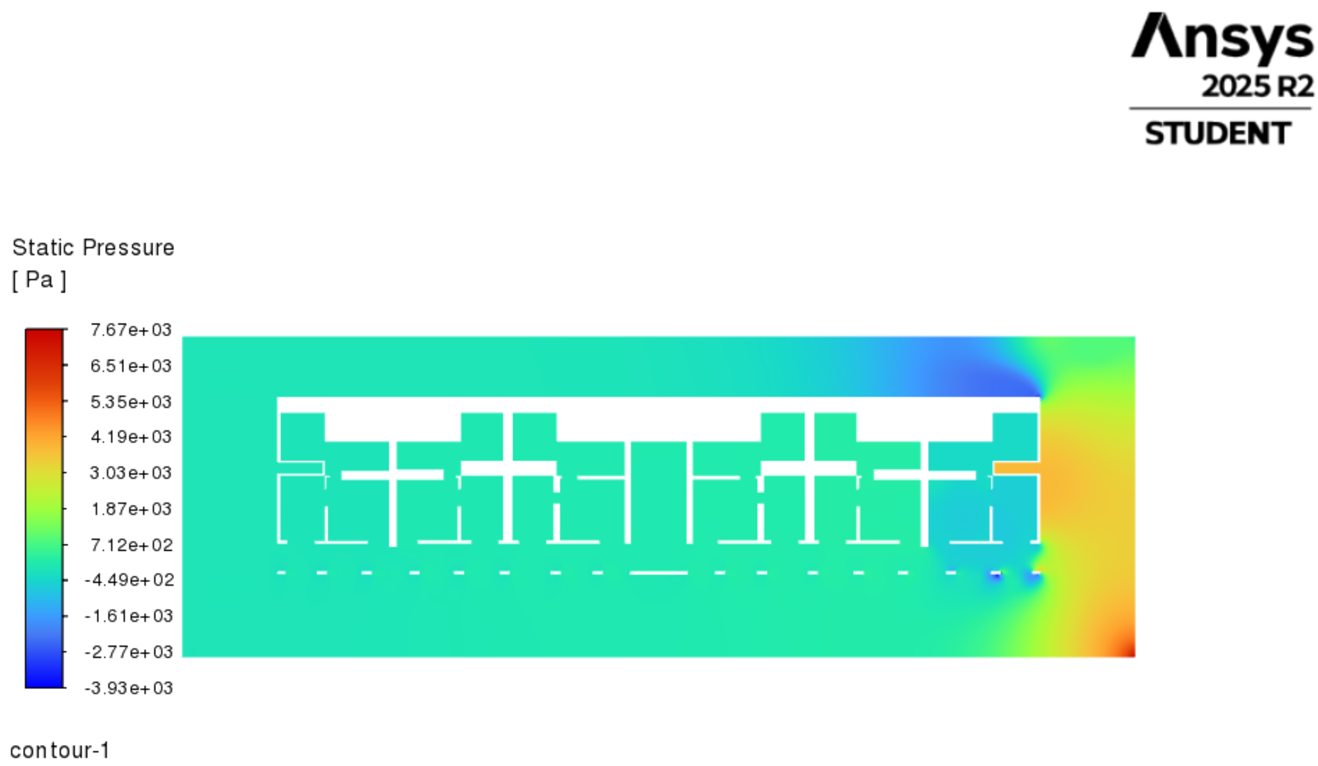


contour-2

contour-1



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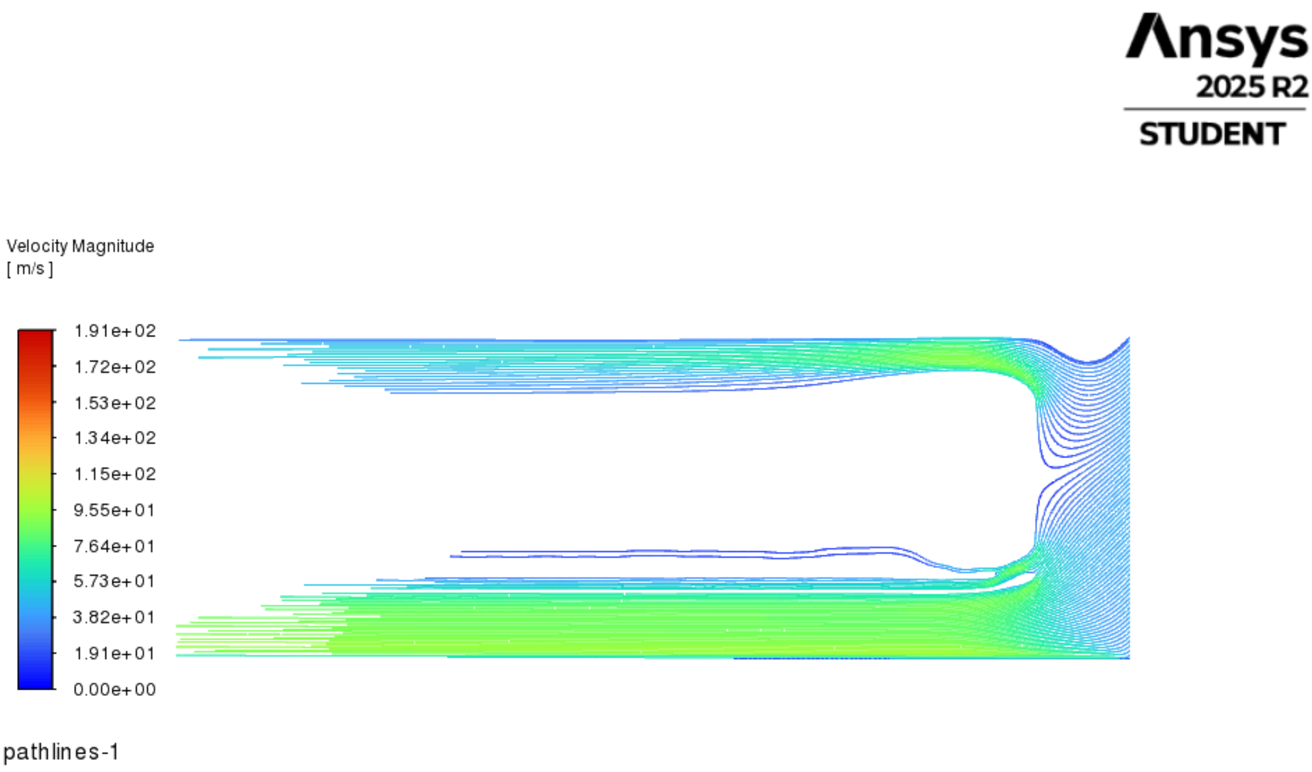
Static Pressure
[Pa]
7.67e+03
6.51e+03
5.35e+03
4.19e+03
3.03e+03
1.87e+03
7.12e+02
-4.49e+02
-1.61e+03
-2.77e+03
-3.93e+03
contour-1

Pathlines

pathlines-1



Scenes

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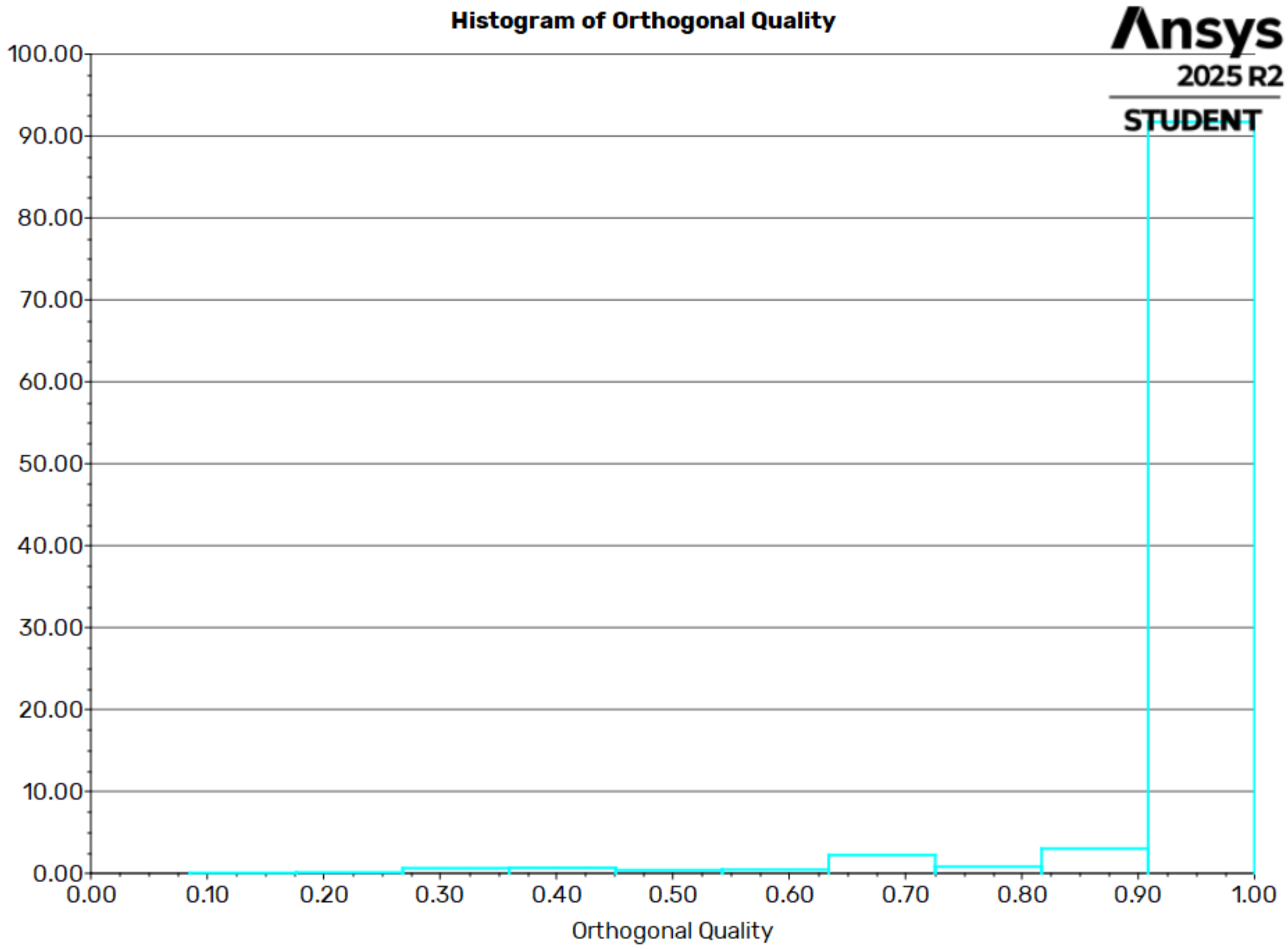

Histogram of Orthogonal Quality
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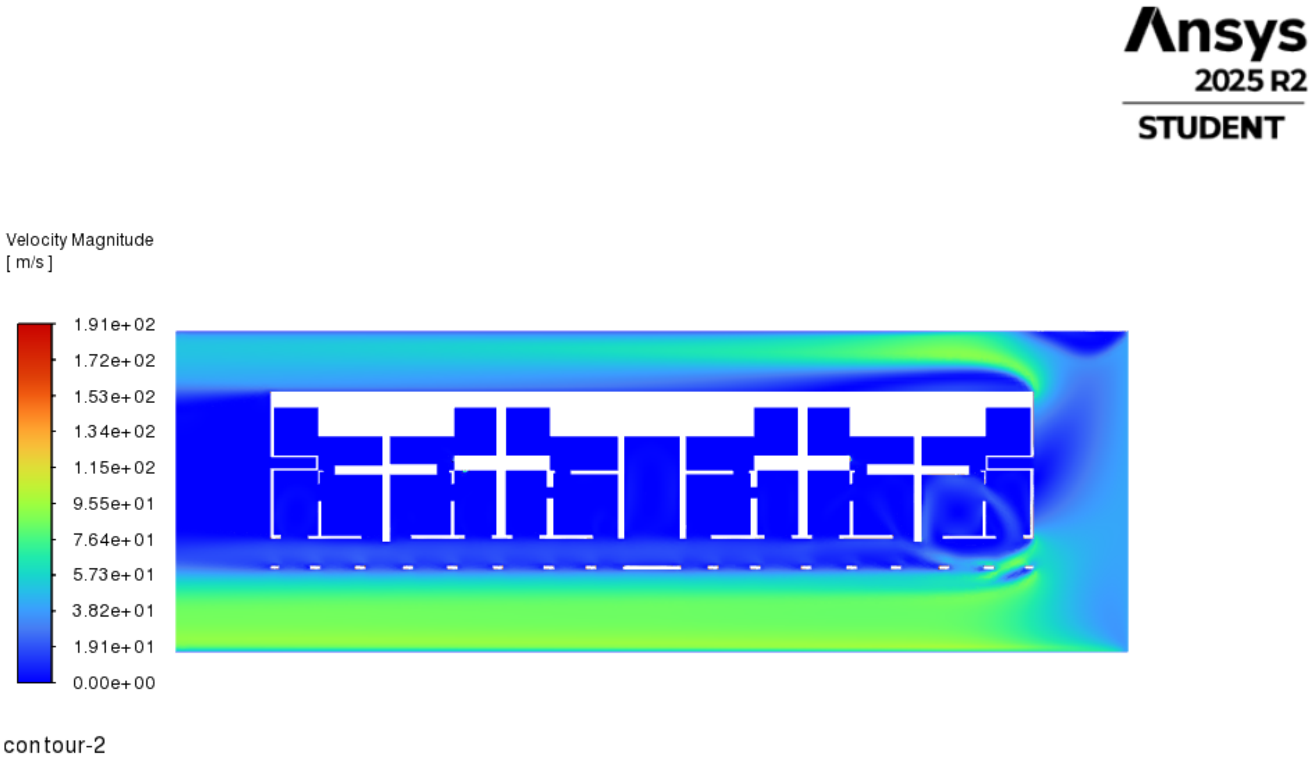
100.00
90.00
80.00
70.00
60.00
50.00
40.00
30.00
20.00
10.00
0.00
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00
Orthogonal Quality



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Velocity Magnitude
[m/s]
1.91e+02
1.72e+02
1.53e+02
1.34e+02
1.15e+02
9.55e+01
7.64e+01
5.73e+01
3.82e+01
1.91e+01
0.00e+00
contour-2

contour-1

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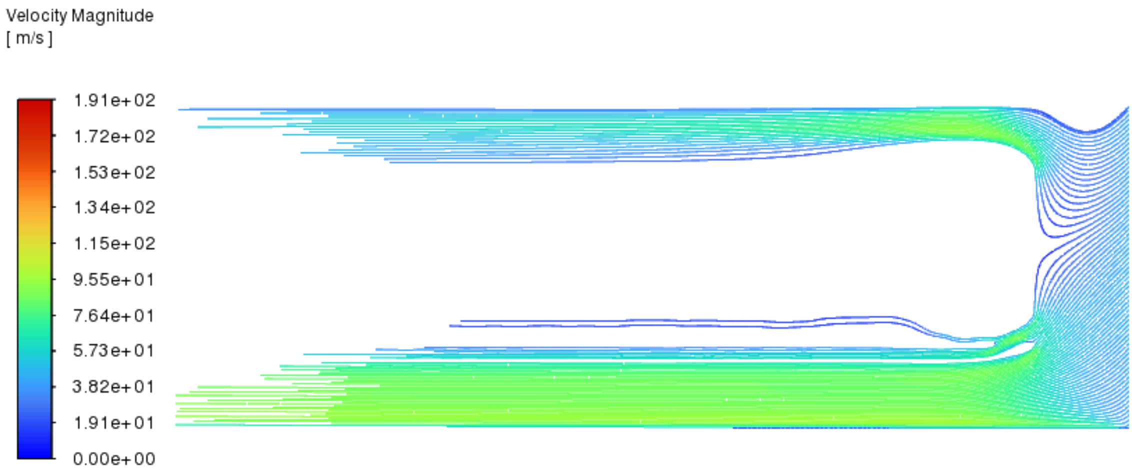

pathlines-1